A FIELD EXTENSION IS REGULAR EXACTLY WHEN IT IS SEPARABLE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Characterisation of separable field extensions, in particular that geometrically reduced extensions are separable. Algebra, Proposition \ref{algebra-proposition-characterize-separable-field-extensions}, cited at proof lines 10495-10497. Gives the direction from regularity to separability.
- A separable field extension is a filtered colimit of smooth algebras over the base. Algebra, Lemma \ref{algebra-lemma-colimit-syntomic}, cited at proof lines 10499-10501. Puts the extension into the shape required by the colimit result.
- A filtered colimit of smooth algebras with Noetherian fibres is a regular ring map. Lemma \ref{lemma-colimit-smooth-regular}, cited at proof lines 10502-10504. Gives regularity of a separable field extension.

Source reference. Original lemma 07EQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9455–9455. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0322: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4005–4005. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07BV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9355–9355. Navigation pointer to a statement the

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Source reference. Original lemma 07EP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9454–9454. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let K/k be a field extension. Then $k \to K$ is a regular ring map if and only if K is a separable field extension of k.*

Proof. If $k \to K$ is regular, then K is geometrically reduced over k, hence K is separable over k by Algebra, Proposition 0322. Conversely, if K/k is separable, then K is a colimit of smooth k-algebras, see Algebra, Lemma 07BV hence is regular by Lemma 07EP. □